



Navigating academic integrity in biomedical research: the impact of large language models on current practices and future directions

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Abstract

As large language models (LLMs) continue to advance, they have garnered widespread public attention and extensive application across numerous industries and academic disciplines. The proliferation of LLMs has sparked considerable research interest, with studies primarily focusing on their technical characteristics and specific application scenarios. However, systematic research examining the impact of LLMs on academic integrity remains relatively scarce. Academic integrity is of paramount importance in the biomedical field. Therefore, this paper aims to examine both the opportunities and challenges that LLMs present to academic integrity in the biomedical field, and proposes solutions for optimizing the beneficial applications of LLMs. From a positive perspective, LLMs offer substantial benefits to researchers by enhancing research efficiency, improving research quality, and facilitating the generation and dissemination of academic insights. However, they also present numerous challenges, including the potential for promoting academic misconduct, generating content inaccuracies or ambiguous expressions, introducing bias and fairness concerns, compromising peer review mechanisms, facilitating the dissemination of misinformation, and undermining higher education – all of which demand careful attention. To address these issues, we propose solutions and feasible strategies centered on ten core dimensions: establishing policies and regulatory guidelines, enhancing AI literacy and application capabilities, developing and improving relevant technical tools, establishing human–AI collaboration models, reforming peer review procedures and academic evaluation systems, promoting international cooperation and standardization, increasing transparency and strengthening disclosure, reinforcing professional ethics education, and advancing artificial intelligence detection technologies. Overall, while LLMs undoubtedly pose challenges for maintaining academic integrity, their potential for positive impact remains promising. It is anticipated that with technological advancement and improved ethical standards, LLMs will ultimately preserve and strengthen academic integrity.

Keywords: academic integrity, academic misconduct, challenges, large language models

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Introduction

Large language models (LLMs), as a type of artificial intelligence (AI) model rooted in natural language processing (NLP), achieve the comprehension and generation of natural language through extensive training on massive datasets. They occupy a pivotal

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role in the field of AI and are profoundly reshaping the landscape of academic research and education^[1]. LLMs exemplify the most recent technological advancements in the domain of NLP. Their fundamental characteristics include a decoder-based architectural design, the pioneering application of transformer models, and the integration of the reinforcement learning from human feedback mechanism^[2]. LLMs find extensive applications in academic contexts, primarily encompassing academic writing assistance, visualization content generation, support for research processes, peer review assistance, and teaching practices. LLMs offer opportunities to promote academic integrity by assisting researchers in improving efficiency and enhancing the quality of their work. However, they also pose challenges related to academic misconduct, accuracy, originality, and ethical considerations.

LLMs have fundamentally transformed academic practices; however, their impact on academic integrity remains inadequately examined. While 68% of researchers believe AI may facilitate plagiarism^[3], comprehensive analyses of this relationship remain scarce. Academic integrity encompasses ethical standards that ensure research authenticity, reliability, and rigor – principles essential for maintaining academic quality and community reputation. LLMs have profoundly influenced traditional academic practices by fostering innovations in academic writing standards, research methodologies, and pedagogical models. Although the initial attitude of the academic community toward the use of LLMs ranged from skepticism to outright dismissal, as LLMs have continued to evolve, their positive value in academic research has been gradually recognized^[4]. At present, LLMs cannot be regarded as authors in the publication of academic work^[5], since doing so would violate copyright laws and academic ethical guidelines; however, they can serve as research assistance tools with both potential and risks^[6].

The increasingly widespread application of LLMs in academic fields has raised critical questions about their impact on academic integrity, yet systematic studies on this subject are still relatively scarce. Existing studies primarily focus on the technical characteristics of LLMs and their specific application scenarios, such as research on the use of ChatGPT in obstetric and gynecological management and diagnosis^[7], investigations into clinical anesthesia practices^[8], applications of machine learning in preclinical drug development, clinical trial management, and data processing^[9], practice of medical education research^[10], and studies on its applications in the prevention and control of the COVID-19 pandemic^[11]. While some studies have highlighted the significant challenges LLMs pose to academic integrity^[12], there remains a lack of comprehensive evaluations and in-depth analyses of their overall impact. Therefore, it is crucial to explore mechanisms to balance the promotion of academic innovation enabled by LLMs with the preservation of academic integrity, ascertain how to leverage their positive potential in advancing scientific research while mitigating potential adverse effects, and identify actionable strategies to protect academic integrity amid their rapid development. The implications of these challenges extend across all medical disciplines, with surgical specialties confronting distinctive considerations due to the precision required in surgical procedures, the critical nature of surgical decision making, and the direct impact on patient outcomes. While this review addresses academic integrity challenges broadly applicable

HIGHLIGHTS

- The increasing adoption and recognition of large language models (LLMs) within the academic community.
- As a significant breakthrough in the field of AI, LLMs have introduced transformative advancements and innovations to academic research.
- The academic community has increasingly acknowledged the potential challenges of LLMs to academic integrity, and the challenges posed by LLMs to academia are becoming increasingly diversified.
- To systematically address the challenges posed by the application of LLMs in the academic domain, a comprehensive governance framework should be established from many dimensions.

across healthcare disciplines, the principles and recommendations presented herein are of particular relevance to surgical research, education, and practice, where accuracy and reliability are paramount.

This review systematically examines the impact of LLMs on academic integrity within the biomedical sciences and provides a comprehensive overview of their primary applications in academia, as well as the opportunities and challenges associated with their use. It first provides an in-depth analysis of the practical applications of LLMs, such as manuscript preparation, data analysis, figure generation, and peer review. It then systematically evaluates the potential value of LLMs in enhancing research efficiency, improving the quality of scientific output, and facilitating the acquisition and dissemination of scholarly knowledge. Moreover, this review highlights the key challenges LLMs encounter in relation to academic integrity, reliability of generated content, originality, fairness, and adherence to ethical standards. Finally, this review proposes systematic strategies across multiple dimensions, including policy-making, education and training, technological advancement, human-AI collaboration, and reform of peer review mechanisms.

The theoretical contribution of this study lies in systematically mapping the current landscape of research on the impact of LLMs on academic integrity within the biomedical sciences, thereby providing a valuable reference for advancing theoretical understanding in related domains. Its practical value is demonstrated through the provision of actionable recommendations for researchers, developers, and policymakers to optimize the benefits of LLMs while safeguarding academic integrity. Furthermore, this study provides critical insights into promoting the structured integration of LLMs into academic practice, thereby advancing academic innovation and progress. This study was conducted without using AI tools in accordance with the TITAN Guidelines 2025^[13].

Applications of LLMs in academic fields

Surveys indicate that approximately 33% of postdoctoral researchers employ LLMs for purposes such as academic writing, programming, and literature management in their research activities^[14]. Related studies indicate that senior students

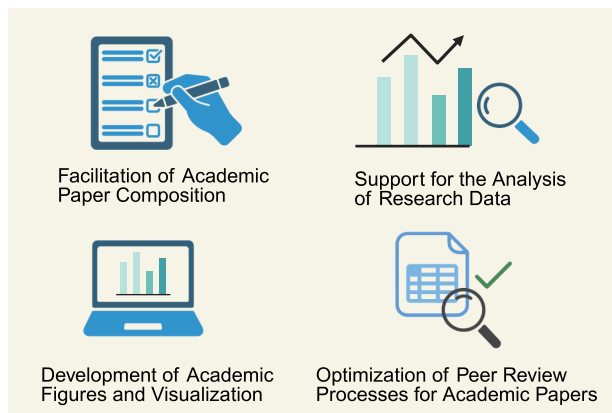


Figure 1. Key applications of large language models (LLMs) in academic research. Large language models are predominantly applied in four major academic domains: (1) Research paper writing assistance: LLMs facilitate manuscript preparation through translation services, comprehensive proof-reading, and multiple perspective analysis, while simultaneously managing auxiliary tasks such as data organization, thereby substantially enhancing research productivity. (2) Research data analysis support: LLMs exhibit emerging capabilities in dataset compilation, analytical processing, and automated code generation, leading to significant improvements in research efficiency. (3) Academic visualization generation: LLMs enhance the traditional visualization process through automated generation of scientific illustrations and data representations, facilitating more effective communication of research findings. (4) Academic peer review support: LLMs contribute to both manuscript self-assessment and formal review processes by providing systematic content evaluation and generating structured review feedback. This figure was created based on the tools provided by Biorender.com (accessed on 20 August 2025).

enrolled in Bachelor of Medicine and Bachelor of Surgery programs – particularly those with strong learning motivation and robust knowledge foundations – exhibit a greater inclination to utilize ChatGPT for academic purposes^[15]. These findings underscore the increasing adoption and recognition of LLMs within the academic community. Their utility is primarily manifested in the following four domains (Fig. 1):

(1) Facilitation of academic paper composition; (2) support for the analysis of research data; (3) development of academic figures and visualization; and (4) optimization of peer review processes for academic papers.

Academic paper writing

LLMs support multiple academic research functions throughout the research workflow. They provide systematic guidance for innovative thinking, assist in hypothesis construction, information retrieval, data analysis, content generation, and reference management^[16–18]. This reduces secondary task interruptions, thereby enabling researchers to focus on core scholarly activities. Upon the completion of research papers, LLMs demonstrate their capability to generate comprehensive and high-quality summaries, particularly excelling at crafting well-structured abstracts^[19]. Empirical studies indicate that research participants commonly regard ChatGPT as significantly enhancing academic research efficiency, with its convenience and productivity being especially evident during the literature review phase^[20]. However, it is important to recognize that excessive reliance on these tools without critical evaluation and human judgment may undermine opportunities for original thinking.

Research data analysis

In the domain of research data analysis, LLMs demonstrate diverse capabilities, including the analysis of datasets, the generation of program code, and the composition of literature reviews. Existing studies suggest that LLMs are capable of systematically analyzing and evaluating multisource textual data using deep learning algorithms, thus offering substantive support to scientific research. The new generation of LLMs, exemplified by ChatGPT, demonstrates exceptional performance in processing large datasets by accurately identifying semantic relationships among texts and revealing latent patterns as well as anomalous features with high precision^[21]. Moreover, LLMs exhibit the capability to translate natural language instructions into executable program code, facilitating the automated generation of diverse code implementations tailored to user requirements^[22]. Within widely adopted research programming environments, including Python and R, LLMs provide assistance in code development while simultaneously identifying and rectifying runtime errors in an intelligent and automated manner^[8]. Research findings further indicate that under specific experimental conditions, LLMs can enable the automation of experimental procedures and intelligent data collection, thereby improving experimental efficiency and strengthening the reproducibility and reliability of research outcomes^[23]. Empirical research additionally corroborates that LLMs, through functionalities such as intelligent data organization, precise sample selection, and comprehensive research process planning, can greatly enhance research efficiency while substantially reducing the time investment required from researchers^[18].

Scientific chart generation

Tables, Figures, and Lists serve as indispensable tools for summarizing and analyzing clinical data; however, conventional manual creation methods are notably time-consuming and demand significant human resources. LLMs provide innovative solutions within this domain, with representative models such as ChatGPT demonstrated to possess the capability of automatically generating sketches and charts^[24,25]. Currently, within the field of scientific chart generation, prominent LLMs beyond ChatGPT include GPT-4 Vision (GPT-4V), Claude, LLaVA, and PaLM 2, among others. Scientific chart generation and data visualization techniques leveraging LLMs play a critical role in academic research, offering robust technical support for the effective visual representation of scholarly findings^[26]. These models have found extensive applications in medical research, including the generation of anatomical joint images for musculoskeletal system studies^[27] and the utilization of GPT-4 V to assist in visualizing the differential induction of genes studied in bone marrow mesenchymal stem cells within multiple myeloma cells^[28]. The recently introduced Data Formulator system integrates natural language input with user interface interaction capabilities, substantially enhancing the efficiency and convenience of generating visualized charts for users^[26].

Academic paper review

LLMs may assist with language editing and structural review during early manuscript screening. However, their ability to assess scientific rigor, methodological soundness, or ethical compliance remains limited and requires oversight by expert human reviewers.

As advanced assistive tools, LLMs may facilitate language editing and structural review during initial manuscript evaluation processes. Deep learning models based on the transformer architecture demonstrate exceptional capabilities in tasks such as literature content extraction, topic identification, and compliance checking of papers by integrating contextual parameters^[29]. From the perspective of authors, LLMs can provide comprehensive and systematic self-assessments of manuscripts and offer specific recommendations for subsequent revisions and optimizations. During the peer review process, LLMs can automate initial screening steps, intelligently identifying and flagging potential issues pertaining to paper quality, academic integrity, and ethical considerations. They can identify systematic biases and methodological limitations in research while reducing plagiarism risks through text similarity analysis^[4,30]. This assists reviewers in objectively evaluating research content from multiple dimensions, thereby aiding in the formulation of more thorough and equitable academic evaluations. LLMs demonstrate significant efficacy in facilitating the drafting of review comments. Studies indicate that article reviews generated by LLMs exhibit a high degree of alignment with the opinions of expert reviewers^[31], providing researchers with high-quality draft reviews and substantially alleviating the

workload associated with drafting literature reviews. This intelligent assistive mechanism significantly enhances the specificity and constructiveness of the review process^[4], while academic publishing efficiency is substantially enhanced through the integration of tools such as ChatGPT to support review and editing tasks^[32]. Furthermore, the application of LLMs in streamlining the peer review process plays a pivotal role in effectively alleviating the workload of reviewers^[33]. Nonetheless, their capacity to assess scientific rigor, methodological validity, or ethical compliance remains limited and requires oversight from expert human reviewers.

The positive impacts of LLMs on academic integrity

As a significant breakthrough in the field of AI, LLMs have introduced transformative advancements and innovations to academic research. In the realm of academic integrity, LLMs have demonstrated a wide range of positive impacts, which can be categorized into three primary areas (Fig. 2): (1) Significantly enhancing research efficiency; (2) effectively improving research quality; and (3) facilitating the acquisition and dissemination of academic knowledge.

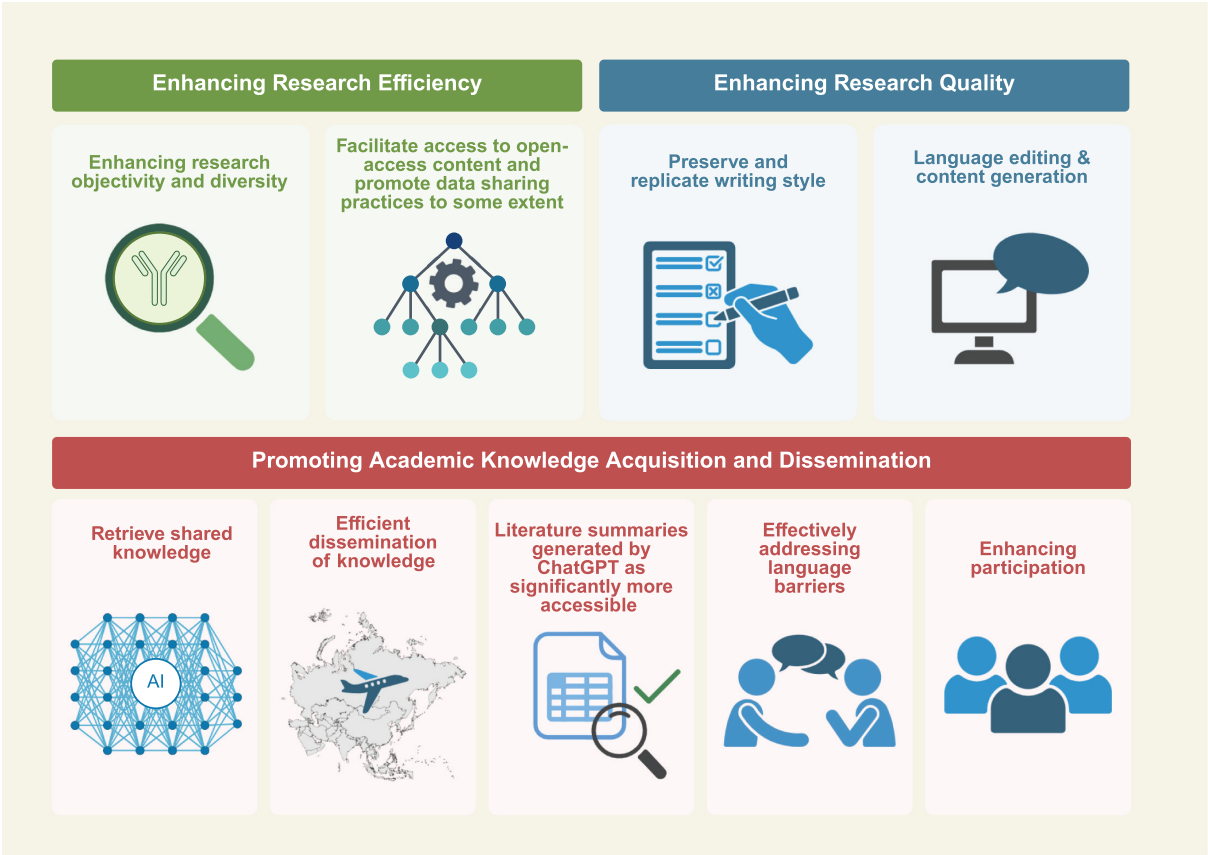


Figure 2. The positive impacts of LLMs on academic integrity. The positive impacts of LLMs on academic integrity manifest in three key aspects: (1) Significant enhancement of research efficiency. LLMs' advanced text generation capabilities expedite writing processes, facilitate outlining, and enhance content development. Furthermore, LLMs streamline the research-informed consent process, thereby conserving valuable research time. (2) Effective improvement of research quality. This enhancement manifests through the provision of high-quality content, encompassing grammar correction and structural optimization, while maintaining author style and elevating the research quality of resource-constrained researchers. (3) Facilitation of academic knowledge acquisition and dissemination. This process enhances information accessibility and promotes efficient knowledge transfer, while effectively removing language barriers to facilitate global knowledge dissemination. This figure was created based on the tools provided by Biorender.com (accessed on 20 August 2025).

Enhancing research efficiency

LLMs can provide diverse analytical perspectives for researchers during the early stages of research, potentially facilitating the identification and mitigation of research biases, thereby enhancing research objectivity and diversity^[34]. Furthermore, LLMs can facilitate access to open-access content and promote data-sharing practices to some extent, enabling researchers to obtain relevant information efficiently. This approach not only fosters global research collaboration and reduces information barriers, thereby improving research efficiency, but also effectively enhances equitable access to knowledge. Studies have shown that, despite certain limitations, employing intelligent dialogue systems to obtain informed consent for research can significantly enhance procedural efficiency and mitigate research delays^[35]. In summary, the multifaceted functionalities of LLMs not only save substantial amounts of research time but also significantly enhance the efficiency of academic research.

Enhancing research quality

LLMs can enhance the language quality, grammar, and readability of research papers, particularly benefiting non-native English speakers. One of the key features of LLMs, particularly regarding writing style, is their ability to preserve and replicate the distinctiveness of an author's text. This characteristic is especially advantageous for individuals with limited writing abilities resulting from functional decline, illness, or advanced age, as it facilitates the preservation of their expressive capabilities. Research indicates that by using a diverse corpus of personal data for fine-tuning, LLMs can effectively replicate the unique writing traits of individual users^[36].

In the field of language editing, LLMs demonstrate remarkable advantages: Texts generated by LLMs typically showcase objective and neutral language features as well as high grammatical accuracy^[37]; they can rectify spelling and grammatical errors, detect inappropriate terminology use, and systematically refine sentence structures^[38–40].

In the realm of content generation, LLMs display outstanding performance. For instance, the AUTOGEN model showcases exceptional text generation capabilities in tasks associated with near transfer learning^[36], offering strong evidence of its superiority in content quality. By sustaining unique writing styles and enhancing content quality, LLMs can greatly elevate the standards of academic writing, thereby empowering researchers to produce more accessible and comprehensible academic outputs. Nevertheless, LLMs cannot independently enhance methodological rigor, conceptual depth, or data validity – all of which represent critical dimensions of research quality.

Promoting academic knowledge acquisition and dissemination

LLMs can be applied within the manufacturing industry to retrieve shared knowledge and provide precise responses to novice inquiries^[41]. In the domain of scientific research, LLMs similarly demonstrate comparable excellence in facilitating the dissemination of knowledge. Studies have shown that these models not only offer accurate guidance to research novices but also accelerate their acquisition of research competencies^[42]. ChatGPT, in particular, has been widely acknowledged for its utility in supporting knowledge transfer, with evidence spanning healthcare interns to seasoned

professionals^[43]. Specifically, studies suggest that approximately 70% of users perceive the literature summaries generated by ChatGPT as significantly more accessible^[44]. By deploying LLMs such as ChatGPT, it becomes possible to build shared knowledge repositories, ensure efficient dissemination of knowledge, and provide accessible knowledge interfaces for non-specialists^[45].

LLMs play a critical role in facilitating global knowledge dissemination, while significantly improving the accessibility and universality of knowledge resources. As an effective tool for overcoming language barriers, LLMs provide academic writing support to non-native English researchers, thereby enhancing their participation and expressive proficiency in international academic communication. This technological breakthrough holds particular significance for researchers in non-English-speaking countries (e.g., Pakistan), effectively addressing language barriers in international academic exchanges and promoting the broader dissemination of global scientific knowledge^[46].

Challenges of LLMs to academic integrity

While LLMs have introduced considerable convenience to academic research, the academic community has increasingly acknowledged their potential challenges to academic integrity. Recent studies indicate that the challenges posed by LLMs to academia are becoming increasingly diversified. This review systematically categorizes these major challenges (Fig. 3), specifically addressing the following: (1) Plagiarism and academic misconduct caused by LLMs; (2) content accuracy and reliability issues for LLMs; (3) the impact of LLMs on academic originality and creativity; (4) fairness and ethical issues of LLMs; (5) academic image fabrication; (6) manipulation of peer review; (7) the intersection of social media, LLMs, and academic integrity; and (8) LLM use in higher education and student assessment.

Plagiarism and academic misconduct caused by LLMs

The academic community is increasingly recognizing the use of content generated by LLMs during the writing process as a novel form of academic misconduct, termed “Algiarism” (AI plagiarism). This practice encompasses using artificial intelligence tools to generate content and submitting or publishing it as one's original work without appropriately acknowledging AI contributions or directly plagiarizing AI outputs^[47]. According to traditional standards for defining plagiarism, the use of text generated by LLMs similarly raises concerns regarding academic norms, especially in relation to verbatim copying of consecutive text (typically defined as five to seven contiguous phrases)^[48]. Even when appropriately cited, such usage may still constitute academic misconduct. The direct copying and use of content generated by ChatGPT not only breaches fundamental principles of academic integrity but also represents a severe violation of intellectual property rights^[49]. Presently, ChatGPT demonstrates substantial limitations in citation accuracy and its capacity to trace scientific reasoning^[50], thereby significantly heightening the risk of unintentional plagiarism. Although certain studies have sought to employ ChatGPT in the development of antiplagiarism tools, these tools have yet to achieve the expected effectiveness^[51]. Furthermore, human evaluators encounter significant challenges in differentiating between human-authored and AI-generated content^[52,53]. Empirical research has revealed that existing AI content detection algorithms demonstrate high error rates in

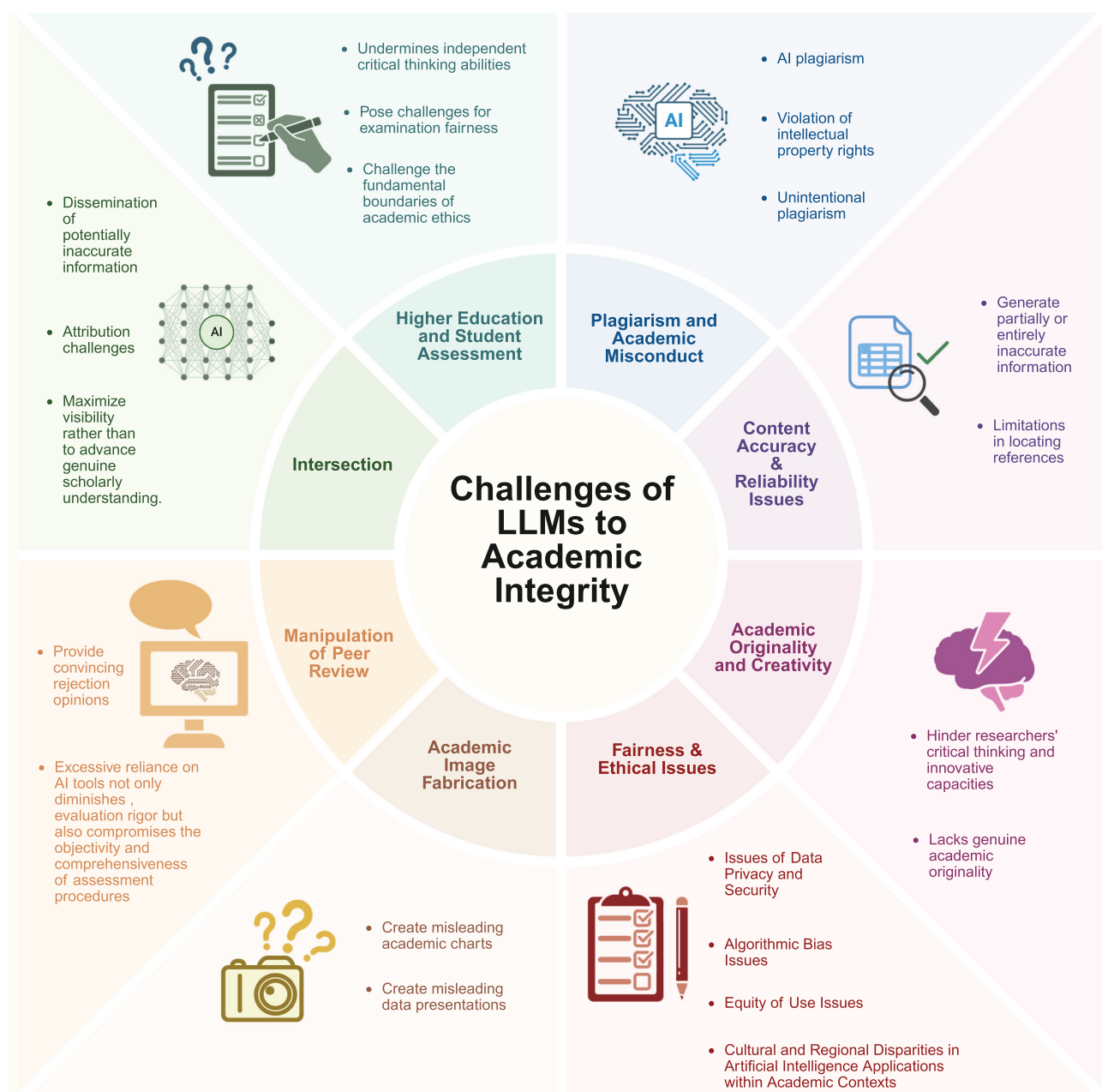


Figure 3. Critical challenges posed by LLMs to academic integrity. Large language models present six fundamental challenges to academic integrity: (1) Plagiarism and academic misconduct: The integration of LLMs introduces elevated risks of inadvertent plagiarism, while the detection of AI-generated content presents significant technical challenges for existing authentication systems. (2) Content accuracy and reliability concerns: LLMs demonstrate limitations in generating consistently accurate content and proper citations. Furthermore, their periodic training data updates create inherent constraints, particularly in rapidly evolving research domains. (3) Impact on academic innovation: LLMs potentially influence researchers' independent critical thinking abilities and creative processes, raising concerns about the preservation of genuine academic originality and intellectual innovation. (4) Ethical and equity considerations: The training data utilized in LLMs raises significant privacy and security concerns, while inherent algorithmic biases potentially compromise equitable access and fair use in academic research. (5) Image generation and authentication: LLM-generated scientific images present unprecedented challenges in verification due to their sophisticated realism, while potentially harboring subtle inaccuracies and increasing risks of data falsification. (6) Peer review system integrity: Over-dependence on LLM-generated peer reviews risks compromising the evaluation process through potential systematic biases and reduced human expert engagement, potentially undermining the fundamental quality assurance mechanisms in academic publishing. (7) The intersection of social media, large language models, and academic integrity: While the profound integration of social media platforms with large language model technologies has facilitated academic advancement, it has simultaneously engendered several challenges, including the rapid dissemination of misinformation, attribution disputes, and content superficiality. (8) Implications for higher education and student assessment: Although LLMs can assist students in mastering knowledge, excessive reliance may weaken students' critical thinking abilities, potentially undermine examination fairness, and challenge the ethical foundations of academia. This figure was created based on the tools provided by Biorender.com (accessed on 20 August 2025).

distinguishing human-authored content from AI-generated material, with nearly 95% of ChatGPT-generated outputs failing to be reliably identified^[54]. Related research suggests that the accuracy rate of professional reviewers in identifying ChatGPT-generated paper abstracts is limited to merely 68%^[55]. Experimental investigations utilizing a double-blind review methodology have demonstrated that even seasoned review experts encounter considerable difficulties in distinguishing AI-generated content from human-authored material^[56,57].

Content accuracy and reliability issues for LLMs

The reliability and validity of academic research critically rely on the accuracy, validity, and authenticity of information^[58]. However, LLMs have exhibited several notable issues. Prior research has revealed that LLMs are prone to the phenomenon termed “hallucination,” which causes them to generate partially or entirely inaccurate information on specific topics^[59]. Moreover, ChatGPT shows marked limitations in locating references^[60] and is often prone to producing false citation details, including fabricated or incorrect DOI and PMID identifiers^[59,61]. Errors in attributing references are frequently observed during the assisted writing process^[62]. In addition to ChatGPT, other AI tools such as BardAI and BingChat also exhibit significant shortcomings in generating citations^[63], suggesting that these tools are not yet reliable for entirely autonomous citation tasks. Improper use of references frequently occurs in AI-generated documents^[64]. Even more alarmingly, AI systems may generate research datasets that appear credible but are, in fact, fabricated, and thereby threaten the foundation of academic integrity. This could not only distort research findings but may also introduce systemic biases in scientific theories and their applications^[65]. Additionally, article abstracts generated by ChatGPT continue to present notable limitations in terms of both their quality and accuracy^[66]. To address these issues, it may be advisable for researchers to consider using quantitative metrics such as recall and precision to assess the effectiveness of such systems^[67]. LLMs face significant limitations regarding knowledge currency. ChatGPT's base model relies on static training data with infrequent updates^[68], which limits its effectiveness in rapidly evolving research fields and poses challenges for cutting-edge applications. This concern is of paramount importance in the field of surgery, where surgical precision and clinical accuracy are essential; therefore, utilizing LLMs for manuscript preparation requires meticulous attention to technical accuracy and clinical relevance. Surgical case reports, technical descriptions, and outcome analyses all demand the highest standards of accuracy, thus rendering the integrity considerations discussed in this article particularly crucial for surgical publications.

The impact of LLMs on academic originality and creativity

LLMs may potentially hinder researchers' critical thinking and innovative capacities. Relevant research surveys suggest that graduate students in the medical field largely acknowledge the pivotal role of critical thinking in academic research^[69]. Given the demonstrated efficiency of LLMs in handling complex tasks, researchers may become overly reliant on these tools, thereby compromising their capacity to cultivate critical and innovative thinking^[70]. It is important to emphasize that LLMs inherently lack authentic critical thinking abilities and cannot engage in independent reasoning or evidence evaluation. Their outputs frequently display mechanized paraphrasing and redundant

expressions, thereby restricting their applicability to creative writing^[71]. LLMs exhibit inherent limitations regarding originality and academic insights. Although discerning the authenticity of their generated content can be challenging^[72], LLMs, such as ChatGPT, may adversely affect the originality of academic research^[73]. These intelligent tools can aid in generating diverse academic documents, including literature reviews, research reports, and even complete academic papers^[17]. However, machine-generated content lacks genuine academic originality and may lead to a proliferation of pseudo-original works, thereby diminishing the value of original research in academia. Specifically, undue reliance on LLMs for drafting research findings and conclusions may significantly compromise the originality of academic contributions^[74].

Fairness and ethical issues of LLMs

Issues of data privacy and security

LLMs present substantial data privacy challenges, as training datasets may contain personally identifiable information from diverse sources, including social media platforms^[75–77]. Privacy breaches could disproportionately affect specific groups, thereby creating significant societal harm. A further critical issue lies in the lack of transparency in LLMs, which impedes effective oversight of decision-making processes, obscures accountability, and complicates governance^[10].

Algorithmic bias issues

The issue of algorithmic bias in LLMs requires thorough and systematic investigation due to its potential to produce discriminatory outcomes^[78]. Algorithmic bias in LLMs primarily manifests in the following key areas: (1) training data bias. If the large-scale datasets used in LLMs' pretraining contain inherent biases, the models are likely to inherit these biases^[79]. When significant biases exist in the representation of specific groups within training datasets, the model may struggle to properly identify and analyze data from other groups. For example, in the medical field, due to the imbalance and quality issues in training data, significant racial disparities have been observed in melanoma recognition tasks^[80]. Research has shown that addressing algorithmic biases can substantially improve the proportion of Black patients receiving effective diagnostic support from 17.7 to 46.5%^[81]. (2) Language usage bias: The inherent ambiguity and context-dependency of language may contribute to biases in semantic understanding^[79]. LLMs may exhibit limitations in handling subtle semantic differences, potentially yielding biased or imprecise outputs. (3) Decision-making bias in algorithms: The decision-making mechanism of LLMs relies on their underlying algorithms, and any embedded biases within these algorithms may directly influence the model's performance^[79]. For instance, when an algorithm overemphasizes particular terms, it may result in systemic biases during the model's generation process.

Equity of use issues

Currently, the development of AI algorithms is predominantly concentrated in high-income countries. Meanwhile, developing and underdeveloped regions encounter notable resource constraints in both accessing and applying AI technologies^[11]. Various factors, such as technological accessibility, levels of digital literacy, cultural disparities, economic inequalities, and

the uneven distribution of educational resources, could collectively intensify the global digital divide^[82].

Cultural and regional disparities in artificial intelligence applications within academic contexts

Artificial intelligence applications in low- and middle-income countries encounter substantial regional barriers, with data sovereignty representing a critical concern. While enhancing data sovereignty proves beneficial for privacy protection, maintaining data sovereignty across organizational, technological, and individual levels remains a significant challenge^[83]. Furthermore, if data exchange is rejected to avoid violations of data sovereignty, such practices may exacerbate the gap between low- and middle-income countries and developed nations, thereby compromising academic equity. The dominant technological frameworks constructed by Western and Global North technology giants may contribute to issues such as the monopolization of computational resources, given that most LLMs are developed by these entities. Moreover, the data used to train LLMs may contain inherent biases, exemplified by ChatGPT's poor performance in certain African languages^[84], which could exacerbate regional disparities. Digital divides and digital colonialism warrant equal consideration, including limitations in elderly populations' use of digital technologies^[85], as well as technological disparities between low- and middle-income countries and developed nations arising from differential levels of technological mastery and varying degrees of access and application, which may further undermine academic equity.

Academic image fabrication

LLMs are capable of altering both the row and column structures of data tables as well as manipulating textual content to create misleading academic charts and data presentations, modifications that are frequently challenging to detect on the surface^[40]. An even more striking observation is that immunofluorescence images and blood smear samples generated by ChatGPT appear so realistic that they could potentially mislead less experienced researchers^[86]. Traditional methodologies for verifying the authenticity of images exhibit substantial limitations in identifying academic images generated by AI. For example, when identifying Western Blot images, researchers frequently rely on band intensity as an indicator to estimate protein concentration. Unfortunately, this method proves inadequate for detecting fabricated images generated by GAN models^[87]. Consequently, while acknowledging the technical advantages posed by LLMs, it is imperative to remain vigilant about potential risks, especially concerning challenges to image authenticity and accuracy. The primary risks involve labeling inaccuracies in data tables, errors in graphic rendering, and redundancies within visual elements^[24]. Of even greater concern is the fact that as generative technologies progress, the authenticity of AI-generated experimental images is improving, necessitating higher levels of scholarly discernment from researchers and aggravating the potential dangers of academic misconduct^[87].

Manipulation of peer review

The application of LLMs and chatbots in academic peer review systems appears to be increasingly prevalent, primarily attributed to time constraints^[88]. A comprehensive analysis of peer

reviews reveals that approximately 7–17% of textual content exhibits characteristics similar to LLM-generated content^[89]. This phenomenon underscores the proliferation of LLMs; however, research examining their impact on academic editorial and peer review processes remains limited^[90].

Recent experimental findings demonstrate that LLMs can provide convincing rejection opinions. They also exhibit adequate performance in refuting unreasonable citation requests. Furthermore, LLM-generated comments are difficult to identify by AI detectors^[91,92]. Although LLMs can generate reasonable comments, excessive reliance on AI systems for generating review opinions without adequate human oversight may lead to biased or even misleading assessment outcomes, including the generation of seemingly reasonable citation requests for irrelevant references^[92].

The integration of LLMs could have far-reaching implications for the decision-making mechanisms of academic journals. Excessive reliance on AI models during critical phases of academic review may undermine the quality assurance systems of scientific publications and compromise the integrity of academic evaluation processes. Excessive reliance on AI tools in academic review and manuscript selection processes not only diminishes evaluation rigor but also compromises the objectivity and comprehensiveness of assessment procedures due to the inherent biases of these tools^[4].

The intersection of social media, LLMs, and academic integrity

The convergence of social media platforms with LLM technologies has created a complex ecosystem that profoundly influences academic integrity through multiple interconnected pathways. Social media platforms increasingly serve as venues for scholarly discourse and preliminary research dissemination, while LLMs facilitate content generation and the translation of complex academic concepts for broader audiences. This intersection offers positive contributions, including democratized access to academic discussions, real-time collaborative peer review, and enhanced public engagement with scientific research. However, it simultaneously presents substantial risks to academic integrity. The speed-driven nature of social media communication can encourage researchers to share LLM-generated content without adequate verification, thereby leading to the rapid dissemination of potentially inaccurate information. Attribution challenges emerge when LLM-assisted insights are shared informally on social platforms, thus creating ambiguity regarding intellectual ownership and original contributions. Furthermore, the metrics-driven environment of social media may incentivize the creation of engaging but superficial LLM-generated content designed to maximize visibility rather than to advance genuine scholarly understanding.

LLM applications in higher education and student assessment

LLMs exert multifaceted impacts on student learning; while LLMs can assist students in knowledge acquisition, they may also compromise academic integrity and diminish independent thinking abilities due to over-reliance^[93]. With respect to examination integrity, since traditional assessment methods struggle to distinguish between students' original work and AI-generated content^[56,57], LLM-facilitated academic dishonesty becomes increasingly covert, thereby compromising assessment validity. More concerning is the rise of AI writing platforms, as these

commercialized services directly challenge the fundamental principles of academic ethics, compelling educational institutions to fundamentally restructure assessment systems – shifting from outcome-based evaluation to process-oriented assessment, for example, incorporating oral defenses and real-time writing verification components.

Standardization strategies for the academic application of LLMs

To systematically address the challenges posed by the application of LLMs in the academic domain, a comprehensive governance framework should be established from the following dimensions (Fig. 4): (1) Developing comprehensive policy frameworks and normative guidelines; (2) enhancing the academic community's awareness and application capacity of AI technologies; (3) developing and improving relevant technical tools; (4) establishing a scientifically robust human-machine collaboration model; (5) reform of the peer review process; (6) reform of the academic evaluation system; (7) international collaboration and standardization; (8) transparency and disclosure requirements; (9) professionalism education as the foundation for academic integrity in the LLM era, and (10) advancing AI detection technologies. By implementing these strategies, the academic value of LLMs can be fully realized while proactively addressing associated risks.

Developing comprehensive policy frameworks and normative guidelines

Establishing clear and robust policy frameworks and normative guidelines is a critical foundation for regulating the application of LLMs in the academic domain. At this stage, although some academic journals have issued guidelines for the application of AI, a significant portion of journals have yet to establish relevant guiding frameworks^[94] (Table 1). In addressing major public health events, while ethical controversies may inevitably arise, it remains necessary to establish comprehensive policy frameworks to ensure data transparency^[95]. To tackle the challenges posed by AI to academic integrity, comprehensive and systematic solutions should be established, such as formulating the ChatGPT, Generated Artificial Intelligence, and Native LLMs Accountability Report and Usage Guidelines (CANGARU)^[96], improving policies for the use of AI-generated outputs^[97], and developing robust scientific governance systems^[98]. These policies should possess broad and universal applicability, covering all aspects of the academic process, including scholarly publishing and peer review^[99].

For the use of AI-assisted content, it is essential to establish a clear and robust disclosure mechanism. Given the inherent complexity of AI algorithms, which could lead to researchers having an inadequate understanding of processing procedures and decision-making mechanisms, it is vital to provide comprehensive methodological and data processing disclosures^[65]. The

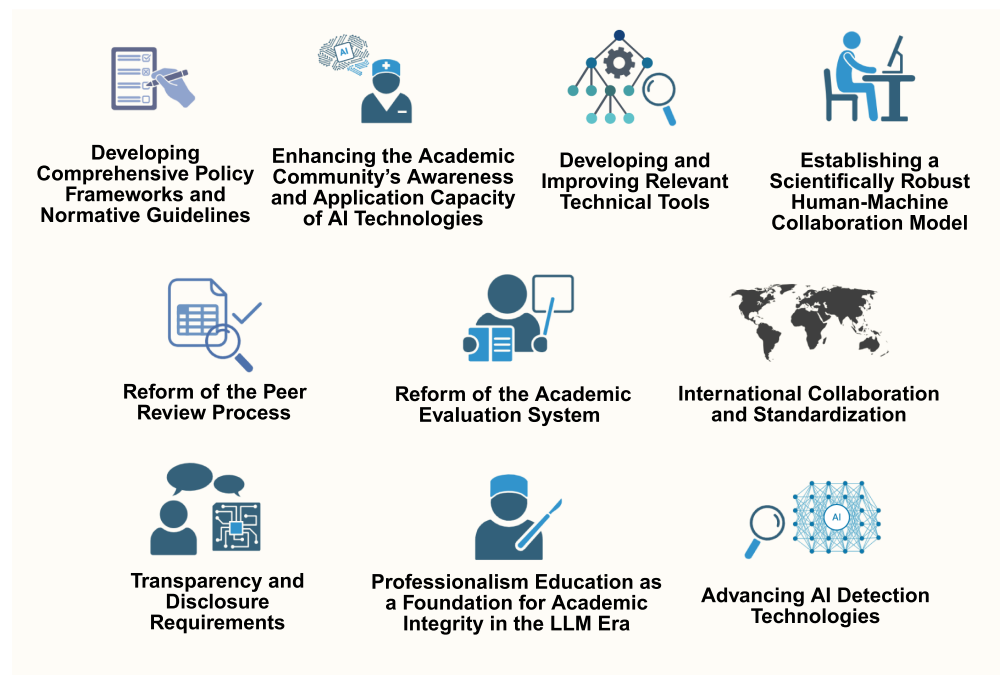


Figure 4. Standardization strategies governing LLMs in academic applications. To systematically mitigate the challenges posed by large language models in academic applications, we propose a comprehensive governance framework encompassing seven critical dimensions: (1) Development and implementation of policy frameworks and regulatory guidelines. (2) Enhancement of the academic community's proficiency in artificial intelligence technologies and their applications. (3) Development and refinement of sophisticated technological tools. (4) Establishment of evidence-based human-machine collaboration frameworks. (5) Enhancement and standardization of academic review processes. (6) Implementation of comprehensive reforms in academic evaluation methodologies. (7) Promotion of international collaboration and standardization initiatives. Implementation of these strategic measures will facilitate the optimal utilization of LLMs' academic potential while systematically addressing associated risks. (8) Enhancement of institutional transparency and implementation of stringent mandatory disclosure requirements. (9) Strengthening of professional ethical standards to safeguard academic integrity. (10) Advancing AI detection technologies. This figure was created based on the tools provided by Biorender.com (accessed on 20 August 2025).

Table 1

Comparative analysis of AI authorship and usage policies across major academic publishers

Publisher/Organization	AI authorship policy	AI usage disclosure requirements	Permitted AI applications	Prohibited AI applications
Nature Portfolio	Prohibited as author; tools only	Mandatory disclosure in methods/acknowledgments	Language editing, formatting assistance	Generative editorial work and autonomous content creation
Elsevier	Not eligible for authorship	Mandatory disclosure	Improve readability and language of the work	Peer review automation, create or alter images
Springer Nature	Does not attribute authorship to AI	Disclosure required in author contribution statement	—	Generative AI images; upload manuscripts into generative AI tools
Wiley	Cannot fulfill authorship requirements	Being transparent with our AI use and development	—	—
Taylor & Francis	Generative AI tools must not be listed as an author	Authors must clearly acknowledge within the article or book any use of Generative AI tools	Idea generation and idea exploration; Language improvement; Interactive online search with LLM-enhanced search engines; Literature classification; Coding assistance	Text or code generation without rigorous revision; synthetic data generation to substitute missing data without robust methodology; generation of any type of content which is inaccurate, including abstracts or supplemental materials; must not upload files, images, or information from unpublished manuscripts

AI, artificial intelligence; LLMs, large language models; NLP, natural language processing.

scope of AI applications must be clearly delineated, permitting its use for background research and literature reviews while strictly prohibiting its application in the analysis of research results^[74]. The use of AI should be restricted to tasks such as language optimization and enhancing readability, rather than to core research activities like data interpretation and conclusion formation. Authors must explicitly disclose their use of AI and take full responsibility for its application^[99]. Specific recommendations include: explicitly detailing the extent of AI involvement, offering comprehensive descriptions of researchers’ oversight during manuscript preparation, and identifying the AI models employed along with their potential biases^[100].

To ensure the effective implementation of the policy framework, we propose the following measures: First, establish a citation standard system for AI-generated content, encompassing clear annotations of the level of AI involvement, ensuring the traceability of original data sources, and safeguarding intellectual property rights through both legal and technical mechanisms. Second, construct a multitiered regulatory framework governing AI usage, establish a dedicated supervisory committee, develop detailed protocols for addressing violations, institute a fair and transparent appeals mechanism, and carry out periodic compliance evaluations. Finally, devise a dynamic adjustment mechanism by systematically evaluating the applicability of existing policies, swiftly addressing challenges introduced by emerging AI technologies, and developing forward-looking policy frameworks through the proactive involvement of diverse stakeholders. These measures are instrumental in ensuring the standardized and sustainable application of AI within the academic sector.

A notable example of comprehensive policy development is the CANGARU (ChatGPT, generative artificial intelligence, and natural large language models for accountable reporting and use guidelines) initiative, which adheres to the EQUATOR Network’s established framework for developing reporting guidelines. This systematic approach employs a multiphase methodology that includes living systematic reviews, bibliometric analysis of existing journal guidelines, and Delphi consensus surveys involving diverse global stakeholders. The

CANGARU framework proposes three distinct criteria lists: (1) a “DON’T” criteria list that outlines prohibited uses across research stages from hypothesis generation to manuscript preparation, (2) disclosure criteria that specify what and how to report LLM usage transparently, and (3) reporting criteria that ensure complete documentation of LLMs when used as research interventions. This evidence-based approach, which involves scientists, clinicians, researchers, statisticians, computer scientists, methodologists, and journal editors from multiple disciplines, exemplifies how comprehensive policy frameworks can be developed through rigorous consensus-building processes. The CANGARU initiative’s emphasis on cross-disciplinary collaboration and international inclusivity provides a valuable model for developing standardized guidelines that can achieve broad adoption across academic institutions and publishing organizations worldwide.

Enhancing the academic community’s awareness and application capacity of AI technologies

Enhancing the academic community’s awareness of and capacity for applying AI technologies is an essential cornerstone for promoting academic integrity. The systematic incorporation of AI literacy into higher education curricula should prioritize fostering students’ critical thinking abilities and their awareness of technological ethics^[101,102]. Medical educators must not only systematically acquire AI-related knowledge but also cultivate professional judgment to evaluate its applicability and feasibility for instructional integration, while thoroughly understanding the developmental opportunities presented by this technological innovation^[103]. University faculty and researchers should undergo professional training on the practical application of AI technologies and their ethical frameworks^[99]; moreover, they should actively update their understanding of the latest advancements and monitor evolving ethical regulations and policies. Educational institutions should establish and continuously refine training frameworks to enhance educators’ professional expertise through diversified and innovative educational approaches. To promote the standardized application of AI

technologies in academia, it is advisable to regularly convene interdisciplinary academic seminars and professional workshops to deeply explore best practices and implementation strategies^[104]. These exchanges should target not only AI developers but also researchers and journal editors within the broader academic community. Simultaneously, modern information technology should be harnessed to build comprehensive online learning resource repositories that consolidate diverse materials, such as medical databases, multimedia platforms, academic journals, and electronic books, thereby offering systematic and practical guidance for the academic community^[104]. Additionally, efficient peer-learning network platforms should be developed to enable in-depth exchanges and sharing of experiences in technological applications, foster innovative thinking through the intersection and integration of ideas, and promote the healthy advancement and standardized adoption of AI technologies in academia.

Development and enhancement of relevant technical tools

Enhancing the interpretability and transparency of LLMs facilitates the supervision of LLM usage processes and reflects the extent of LLM involvement in academic research and scholarly activities. Such transparency measures are conducive to preventing the use of LLMs for academic misconduct. These measures can also encourage researchers to view LLMs critically, engage in independent thinking and verification of their generated content, and use LLMs as auxiliary instruments rather than becoming overly reliant on them. Continuous improvement of existing LLMs through iterative updates, aimed at eliminating biases – whether inherent in the system itself or arising from training data – enhances their intelligence and capability to deliver reliable and accurate outputs to users. If biases can be thoroughly eliminated, such improvements would not only make LLM-generated content more objective and impartial but also significantly advance academic integrity. Furthermore, training LLMs with massive datasets enables them to cover as much of human knowledge as possible, while simultaneously eliminating their “hallucination risks” through algorithmic improvements. Through these enhancements, LLM-generated content would be more reliable and authentic, and the applications of these models would become more extensive. Establishing rigorous scientific conceptual frameworks^[105] is crucial for evaluating the performance and reliability of AI systems and providing clear guidance for their improvement. The synergistic development of these technologies not only effectively curbs academic misconduct but also robustly safeguards academic integrity.

Establishing a scientifically robust human–machine collaboration model

Establishing a scientifically robust human–machine collaboration model is a fundamental safeguard for balancing the application of AI technology and preserving academic integrity. The primary objective is to delineate AI’s supportive role in academic research while underscoring the centrality and decision-making authority of human researchers throughout the research process. To ensure the reliability and originality of academic outputs, it is imperative to institute a systematic review mechanism for AI-assisted content generation while concurrently developing intelligent interactive research platforms to enhance the seamless integration of human–machine collaboration. At the practical

level, developing comprehensive guidelines for human–machine collaboration is essential for standardizing the application of AI technologies within the research process. These guidelines must explicitly delineate the scope of AI tools, operational procedures, and ethical boundaries. Furthermore, actively exploring innovative paradigms of AI–human expert collaboration can fully capitalize on AI’s strengths in data processing and pattern recognition while leveraging human expertise in creative problem-solving and intricate judgments^[106,107]. This synergistic relationship is instrumental in maintaining the integrity and credibility of academic research.

Reform of the peer review process

Reforming the traditional peer review system to effectively address the novel challenges introduced by advancements in AI technology has become a critical issue in academic publishing. In the initial review stage, the incorporation of AI-assisted prescreening mechanisms can greatly improve both the efficiency and accuracy of manuscript selection, with intelligent tools designed for literature similarity detection^[99] serving as a pivotal method to enhance review precision and effectiveness. Developing a hybrid peer review model that integrates human expertise with AI assistance not only capitalizes on the strengths of intelligent review management systems in optimizing processes and resource allocation but also maintains the rigor and integrity of the review process. Moreover, enhancing the transparency of the review process and explicitly disclosing the application of AI tools can significantly bolster the credibility of academic peer review. Furthermore, implementing a system to assess AI literacy among reviewers is crucial for maintaining the quality and reliability of the peer review process. Notably, harnessing the advantages of blockchain technology in data sharing, collection, and management^[36], exploring the development of a decentralized review system may provide innovative pathways to strengthen the credibility of academic peer review. This transformative model is anticipated to redefine the academic review ecosystem in the future.

Reform of the academic evaluation system

Restructuring the existing academic evaluation system in response to the paradigm shift driven by the deep integration of AI technologies in academic research has become a key issue facing the academic community today. This restructuring is primarily reflected in the development of new evaluation standards for academic outputs, which must comprehensively consider the distinctive characteristics of AI-assisted research. To achieve this goal, a tiered evaluation system for the extent of AI involvement should be established, encompassing different levels such as text proofreading, content reorganization, and initial draft drafting^[100], while incorporating researchers’ actual contributions to devise a diversified mechanism for assessing academic contributions. This mechanism should not only balance quantitative metrics and qualitative evaluation but also utilize AI-powered tools for assessing academic impact, providing a more comprehensive and objective basis for evaluating research outputs. On this basis, exploring academic evaluation models grounded in the principles of open science can help foster knowledge sharing and academic collaboration. To ensure the fairness and comprehensiveness of the evaluation process, it is particularly important to establish an evaluation committee

consisting of experts from diverse disciplines. This will facilitate a multidimensional and multifaceted assessment of academic outputs, ultimately leading to the establishment of a more scientific and equitable academic evaluation system.

International collaboration and standardization

Against the backdrop of rapid advancements in AI technologies, it has become imperative to strengthen international collaboration and establish unified standards and norms to address the global challenges posed by AI. Currently, the international community has not yet reached a unified consensus on ethical principles for the use of AI^[108], with significant disparities evident in the policies and practices across different countries and regions^[109]. These disparities can be attributed to a combination of factors, such as uneven technological development, differences in legal frameworks, and the diversity of cultural values. Strengthening international cooperation not only contributes to addressing global challenges effectively but also facilitates the creation of a global research integrity system, advances the development of unified standards and norms, and offers a robust platform for the exchange of experiences^[65]. To this end, the international community should adopt a multi-faceted strategic approach: hosting international academic conferences to foster consensus, forming transnational research teams to develop unified standards, promoting the creation of a global AI academic ethics committee to coordinate national policies, establishing an international database on AI academic integrity for knowledge sharing, and facilitating the global exchange of AI technologies and academic resources to mitigate regional disparities. Implementing these initiatives will foster the development of a more equitable and unified global AI academic ecosystem^[110,111].

Transparency and disclosure requirements

The implementation of mandatory disclosure requirements represents a fundamental component of responsible LLM integration within academic practice. Comprehensive transparency frameworks should encompass several critical dimensions: (1) Technical disclosure, specifying which LLM systems were utilized, their versions, and the specific tasks for which they were employed; (2) process transparency, detailing how LLM outputs were integrated into the research workflow, including verification procedures and human oversight mechanisms; (3) scope delineation, clearly distinguishing the boundaries between LLM assistance and human intellectual contribution; and (4) quality assurance documentation, describing the methods used to verify the accuracy and appropriateness of AI-generated content.

Disclosure statements should be positioned prominently within manuscripts, preferably in both the acknowledgments section and the methodology section, where applicable. For review articles such as the present work, transparency becomes particularly crucial given the potential for LLMs to influence literature synthesis, argumentative structure, and conceptual organization. The academic community must establish standardized disclosure formats that facilitate consistent reporting across disciplines while avoiding both under-disclosure and excessive technical details that obscure rather than clarify the nature of AI assistance.

Furthermore, journals should implement clear editorial policies requiring such disclosures and provide guidance to authors on acceptable levels and types of LLM assistance. This approach

not only maintains academic integrity but also contributes to the broader understanding of how AI tools are being integrated into scholarly practice, thereby informing future policy development and best practices.

Professionalism education as the foundation for academic integrity in the LLM era

In the era of rapid LLM technological advancement, academic research faces unprecedented ethical challenges. Tools such as ChatGPT facilitate text generation, but have simultaneously given rise to new forms of academic misconduct, including “AI ghostwriting” and data fabrication. Against this backdrop, drawing upon the systematic professionalism cultivation model from medical education to construct an academic integrity education system adapted to the artificial intelligence era is essential.

Professionalism emphasizes three core values: Ethical responsibility, academic honesty, and prioritization of public interest. Its cultivation approach holds particular significance for academic standards in the LLM era. First, through human–AI collaborative research ethics courses, students can be guided to appropriately define the boundaries of AI tool usage, for example, permitting literature search assistance in systematic reviews while strictly prohibiting direct generation of research conclusions. Second, establishing algorithm transparency training modules that require students to annotate AI contributions when using LLMs represents an approach similar to conflict of interest declarations in medical research. More importantly, the cultivation of professionalism can help students develop ethical decision-making capabilities, enabling them to uphold academic integrity when faced with temptations such as using AI to manipulate data to increase publication likelihood.

The practice of training professional ethics through standardized patients (SP) in medical schools warrants emulation. Similarly, AI-assisted manuscript review simulation experiments can be designed in which students assume the roles of authors and reviewers to directly experience the damage that academic misconduct inflicts on knowledge systems. This experiential learning approach may be more effective in cultivating internal restraint than punitive measures alone.

As technological transformation reshapes the academic ecosystem, professionalism education provides an ethical foundation for academic integrity development. It not only regulates technology use but also enables the academic community to maintain self-purification capabilities in the AI era through shaping researchers’ professional identity. This may represent the most essential safeguard of human wisdom that academia should uphold amid the rapid advancement of large language models^[112,113].

Advancing AI detection technologies

Academic plagiarism detection systems can be categorized into two main types based on the scope of detection: internal plagiarism detection and external plagiarism detection^[114]. Internal plagiarism detection primarily assesses the consistency of authorship within a single text, aiming to identify self-repeated content without proper citation and attribution, thereby ensuring that appropriate credit is given. External plagiarism detection, on the other hand, focuses on comparing a text with external documents to uncover the misuse of others’ original works, intellectual ideas, and creative expressions^[115]. To

address these behaviors that undermine academic integrity, tailored algorithmic models need to be developed to detect idea plagiarism^[116] and to expand the scope of detection to include cross-lingual plagiarism^[117]. Notably, ensemble learning techniques have proven to be significantly effective in identifying content generated with AI assistance^[118].

In the context of rapid advancements in AI technology, developing and refining relevant technical tools represents a critical strategy for effectively addressing the challenges posed by AI to academic integrity. The primary task is to develop more accurate AI-based content recognition tools for academic purposes to facilitate the identification and analysis of AI-generated content. This requires sustained investment in algorithmic models and analytical tools for AI-generated content^[99], including enhancing models for detecting graphical integrity^[119] and developing detectors for improper model behavior^[120]. While current AI detection tools face significant limitations, emerging technological approaches offer promising avenues for more effective identification of AI-generated content. Watermarking techniques represent a proactive solution whereby imperceptible signatures are embedded during the text generation process, enabling reliable detection even after minor modifications. Advanced stylometric analysis leverages sophisticated linguistic patterns and distinctive writing fingerprints that differentiate human-authored from AI-generated text, transcending simple statistical measures to capture deeper semantic and syntactic characteristics. Citation pattern analysis presents another innovative approach, as AI-generated content frequently exhibits distinctive referencing behaviors and knowledge integration patterns that diverge from conventional human scholarly writing practices. Paradoxically, AI itself is becoming instrumental in detecting AI-generated content, with machine learning models specifically trained to identify subtle artifacts and inconsistencies inherent in synthetic text generation. The convergence of these technologies, in conjunction with block-chain-based verification systems and real-time authentication protocols, suggests a future wherein detection capabilities will significantly surpass current limitations. However, the ultimate success of these technologies will depend upon their integration into comprehensive frameworks that balance detection accuracy with practical implementation considerations, thereby ensuring they serve as effective safeguards rather than impediments to legitimate academic innovation.

The influence of LLMs on academic integrity is inherently dual-faceted, presenting opportunities to enhance academic efficiency and foster innovative research methodologies, while concurrently introducing challenges to academic integrity and quality control. Addressing these multifaceted impacts necessitates collaborative efforts among diverse stakeholders, including academia, technology developers, and policymakers, achieved through the formulation of comprehensive management strategies and technical guidelines to fully harness the positive potential of LLMs while effectively mitigating associated risks. Looking forward, research should explore the long-term implications of AI-generated content on the academic ecosystem, continuously refine associated ethical frameworks and evaluation systems, and improve both the anticipation of and responses to emerging technological application scenarios. Furthermore, the establishment of robust regulatory mechanisms, improvement in academic evaluation systems, and reinforcement of international collaboration will be critical safeguards to ensure the sustainable and healthy development of LLMs.

Through multidimensional and multilevel collaborative initiatives, the academic community can seek to fully unlock the positive potential of LLMs in academic research while upholding academic integrity, thus promoting academic innovation and sustainable development.

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Consent

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Author contributions

A.L., Z.C., A.J., and B.T.: Writing – original draft, investigation, writing – review and editing, visualization. C.Q., L.Z., W.M., D.Z., W.G., M.X., G.C., S.P., H.Z.H.W., L.Z., H.Z., X.D., and Q.C.: Writing – review and editing. J.Z. and P.L.: Writing – original draft, conceptualization, writing – review and editing.

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